A Project Report on

Diabetes Prediction Using Data Analytics and Machine Learning Algorithms

Submitted in partial fulfillment for award of

Bachelor of Technology

Degree

in

Computer Science and Engineering

By

J.Kalyani (Y22ACS465) **H.Likitha (Y22ACS463)**

M.Karthik (Y22ACS495) **M.Sujata (L23ACS609)**

Under the guidance of

V . Naveen Kumar, M.Tech.,(Ph.D.)

Assistant Professor

Department of Computer Science and Engineering

Bapatla Engineering College

(Autonomous)

(Affiliated to Acharya Nagarjuna University)

BAPATLA – 522 102, Andhra Pradesh, INDIA

2025-2026

Department of
Computer Science and Engineering

CERTIFICATE

This is to certify that the project report entitled **Diabetes Prediction Using Data Analytics and Machine Learning Algorithms** that is being submitted by J.Kalyani (Y22ACS465), H.Likitha (Y22ACS463), M.Karthik (Y22ACS495), M.Sujata (L23ACS609) in partial fulfillment for the award of the Degree of Bachelor of Technology in Computer Science & Engineering to the Acharya Nagarjuna University is a record of bonafide work carried out by them under our guidance and supervision.

Date:

Signature of the Guide	**Signature of the HOD**
V. Naveen Kumar	**Dr . M. Rajesh Babu**
Assistant Professor	**Assoc. Prof. & Head**

DECLARATION

We declare that this project work is composed by ourselves, that the work contained herein is our own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

J . Kalyani (Y22ACS465)
H . Likitha (Y22ACS463)
M . Karthik (Y22ACS495)
M . Sujata (L23ACS609)

Acknowledgement

We sincerely thank the following distinguished personalities who have offered their valuable advice and support for the successful completion of this work.

We are deeply indebted to our most respected guide, **V.Naveen Kumar, Assistant Professor**, Department of Computer Science and Engineering, for his/her valuable guidance, constant encouragement, constructive suggestions, and inspiring support throughout the course of this project.

We extend our sincere thanks to **Dr. M. Rajesh Babu, Associate Professor & Head of the Department**, Department of Computer Science and Engineering, for his cooperation and for providing the necessary facilities and resources to carry out this work successfully.

We would also like to express our heartfelt gratitude to our beloved **Principal, Dr. B. Srinivasarao**, for providing the required infrastructure, online resources, and institutional support to complete this project.

We express our sincere thanks to our **Project Coordinators, Dr. C. Prakasa Rao, Professor, Department of CSE, and Dr. P. Venkata Lakshmi, Associate Professor, Department of CSE**, for their valuable suggestions and guidance in preparing and presenting this document.

We are always thankful to our **parents and family members** for their continuous support, encouragement, and motivation. Without their support, we would not have been able to complete this project work successfully.

Finally, we extend our sincere gratitude to all the teaching faculty and non-teaching staff of the Department of Computer Science and Engineering who helped us directly or indirectly with their cooperation and encouragement.

J.Kalyani (Y22ACS465)

H.Likitha (Y22ACS463)

M.Karthik (Y22ACS495)

M.Sujata (L23ACS609)

ABSTRACT

Diabetes mellitus is one of the fastest-growing chronic diseases globally and poses a major challenge to public healthcare systems. Early and accurate prediction of diabetes risk is essential for enabling timely medical intervention, lifestyle modifications, and preventive care. This project presents a Diabetes Risk Predictor, an explainable machine learning-based application developed using the Streamlit framework, which allows clinicians and patients to assess diabetes risk based on important clinical and demographic parameters.

The system is built using a diabetes prediction dataset containing approximately 100,000 patient records with features such as age, Body Mass Index (BMI), HbA1c level, blood glucose level, hypertension, heart disease, gender, and smoking history. Multiple classification algorithms including Logistic Regression, Decision Tree, Random Forest, and XGBoost were trained and evaluated. Based on comparative performance analysis, the best-performing model was automatically selected and saved for deployment.

A major contribution of this project is the integration of SHAP (SHapley Additive exPlanations) to provide both global-level and patient-level interpretability of model predictions. The interactive dashboard includes modules for Global Feature Importance, SHAP-based Explanation, Clinical Summary, and Personalized Recommendations. The system classifies patients into HIGH, MEDIUM, and LOW risk categories and generates dynamic clinical interpretations along with customized health and dietary recommendations.

The project demonstrates the practical applicability of Explainable Artificial Intelligence (XAI) in healthcare by combining predictive accuracy with transparency and decision support, thereby making the system useful for early diabetes risk assessment and clinical assistance.

KEY WORDS:

Diabetes Prediction, Machine Learning (ML), Explainable Artificial Intelligence (XAI), SHAP, Predictive Modeling, Healthcare Analytics, Risk Assessment, Streamlit Dashboard.

Table of Contents

ABSTRACT iv

List of Figures vii

List of Tables viii

1 INTRODUCTION 1

1.1 MOTIVATION 2

1.2 PROBLEM DEFINITION 2

1.3 OBJECTIVE OF PROJECT 3

2 LITERATURE SURVEY 5

2.1 EXISTING SYSTEM 5

2.2 DISADVANTAGES OF EXISTING SYSTEM 7

2.3 PROPOSED SYSTEM 8

2.4 ADVANTAGES OF PROPOSED SYSTEM 9

3 ANALYSIS 11

3.1 SOFTWARE REQUIREMENT SPECIFICATION 11

3.1.1 User Requirements 11

3.1.2 Software Requirements 12

3.1.3 Hardware Requirements 13

3.2 CONTENT DIAGRAM OF PROJECT 15

4 DESIGN 17

4.1 INTRODUCTION 17

4.2 DFD/ ER/ UML DIAGRAM 18

4.2.1 Data Flow Diagram (DFD) 19

4.2.2 Entity-Relationship (ER) Diagram 20

4.2.3 Use Case Diagram 22

4.2.4 Sequence Diagram 24

4.2.5 Class Diagram 26

4.3 DATA DICTIONARY 28

4.3.1 Input Data Dictionary 28
4.3.2 Output Data Dictionary 29
5 IMPLEMENTATION & RESULTS 30
5.1 METHOD OF IMPLEMENTATION 30
5.1.1 Implementation of Algorithms 31
5.1.2 Output Screens 35
5.1.3 Result Analysis 39
6 TESTING & VALIDATION 41
6.1 FEATURES TO BE TESTED 41
6.2 FEATURES NOT TO BE TESTED 42
6.3 TESTING TOOLS AND ENVIRONMENT 43
6.4 DESIGN OF TEST CASES AND SCENARIOS 44
7 CONCLUSION & FUTURE ENHANCEMENT 49
8 BIBLIOGRAPHY 50

List of Figures

Fig 3.1 Content Diagram.....15
Fig 4.1 Data Flow Diagram 20
Fig 4.2 Entity-Relationship Diagram.....22
Fig 4.3 Use Case Diagram 24
Fig 4.4 Sequence Diagram.....26
Fig 4.5 Class Diagram.....28
Fig 5.1 Home Page.....35
Fig 5.2 User Input Interface 36
Fig 5.3 Prediction Result Screen.....36
Fig 5.4 Visualization Output Screen 37
Fig 5.5 Clinical Summary Screen 38
Fig 5.6 Recommendation Screen 39
Fig 6.1 Test Case 1.....45
Fig 6.2 Test Case 2.....45
Fig 6.3 Test Case 3.....46
Fig 6.4 Test Case 4.....46
Fig 6.5 Test Case 5.....47
Fig 6.6 Test Case 6.....47
Fig 6.7 Test Case 7.....48
Fig 6.8 Test Case 8.....48

List of Tables

Table 3.1 Software Requirements 12
Table 3.2 Hardware Requirements 14
Table 4.1 Input Data Dictionary 29
Table 4.2 Output Data Dictionary 29

1 INTRODUCTION

Diabetes mellitus is one of the most common chronic diseases worldwide and has become a major public health concern. According to the International Diabetes Federation (IDF), approximately 537 million adults worldwide are living with diabetes, and this number is expected to rise significantly in the coming years. In India alone, over 80 million individuals are affected, making early detection and preventive care highly important [9]. It is a metabolic disorder caused by inadequate insulin production or ineffective insulin utilization in the body, resulting in elevated blood glucose levels. If not identified and managed at an early stage, diabetes can lead to severe complications such as cardiovascular diseases, kidney failure, nerve damage, and vision loss [10].

With the increasing availability of healthcare data, machine learning has emerged as a valuable tool for disease prediction and clinical decision support. By analyzing patient information such as age, body mass index (BMI), HbA1c level, blood glucose level, hypertension, heart disease, gender, and smoking history, machine learning models can help identify individuals who may be at risk of developing diabetes [1], [2].

Although machine learning models can provide accurate predictions, many of them lack interpretability and behave as black-box systems. In healthcare applications, prediction alone is not sufficient; both clinicians and patients need to understand the reasons behind the result. Therefore, explainable and transparent predictive systems are essential for building trust and improving real-world usability.

This project presents an interactive diabetes prediction system that combines data analytics, machine learning, and explainable artificial intelligence to estimate diabetes risk and provide meaningful insights. The system predicts patient risk levels, explains the impact of important features, and generates clinical interpretation along with personalized health and dietary recommendations through a user-friendly Streamlit dashboard.

1.1 MOTIVATION

The motivation behind this project arises from the growing need for early and accessible diabetes risk assessment. In many cases, diabetes remains undetected until noticeable symptoms or complications appear, which reduces the effectiveness of preventive care. Early prediction can help individuals take timely medical advice and lifestyle modifications before the condition worsens.

Traditional clinical diagnosis of diabetes mainly relies on laboratory tests such as fasting blood glucose and HbA1c measurements. While these methods are medically reliable, they are generally used after symptoms or risk indicators become noticeable. Risk stratification and early prediction using patient health profiles remain underutilized in many healthcare settings.

Another major challenge in current predictive healthcare systems is the lack of interpretability. Many machine learning models can provide accurate predictions, but they do not explain why a patient is classified as high or low risk. In clinical environments, this lack of transparency can reduce trust and limit practical adoption.

Therefore, this project was motivated by the need to create a clinically useful, transparent, and interactive tool that can assist both healthcare providers and patients in understanding diabetes risk, its contributing factors, and possible preventive interventions.

1.2 PROBLEM DEFINITION

The increasing number of diabetes cases worldwide highlights the need for efficient and intelligent risk prediction systems. Although machine learning has been widely applied for disease prediction, many existing systems focus only on generating prediction outputs without providing explanation or practical usability for healthcare environments.

The major problems identified in existing diabetes prediction systems are as follows:

- Most systems provide only a binary prediction (diabetic or non-diabetic) without indicating the degree of risk.
- Many machine learning models operate as black boxes, making it difficult to understand why a prediction was made.
- Existing solutions often lack interactive dashboards for real-time patient input and analysis.
- They do not usually provide clinical interpretation or personalized health recommendations based on the patient's condition.
- The absence of explainability reduces trust and usability in real-world healthcare applications.

To overcome these limitations, this project proposes a system that predicts diabetes probability, categorizes patient risk into meaningful levels, explains the contribution of each feature using SHAP, and provides health and dietary recommendations through an interactive dashboard.

1.3 OBJECTIVE OF PROJECT

The main objective of this project is to develop an Explainable Machine Learning-Based Diabetes Risk Prediction System that can accurately predict the likelihood of diabetes using patient clinical and demographic information while also providing interpretable results for better healthcare decision support.

The specific objectives of the project are as follows:

- To collect and preprocess diabetes-related patient data by handling categorical features, numerical attributes, and scaling techniques for effective model training.
- To build and compare multiple machine learning classification models such as Logistic Regression, Decision Tree, Random Forest, and XGBoost for diabetes risk prediction.
- To select the best-performing model and use it to predict the probability of diabetes based on user input parameters.
- To categorize the predicted diabetes probability into meaningful risk levels such as LOW, MEDIUM, and HIGH for better clinical interpretation and patient assessment.

- To integrate SHAP (SHapley Additive exPlanations) for identifying the most influential features and improving the interpretability of the prediction results.
- To develop an interactive Streamlit dashboard that provides real-time diabetes prediction, clinical interpretation, and personalized health, lifestyle, and dietary recommendations.

2 LITERATURE SURVEY

The application of Machine Learning (ML) in the healthcare domain has gained significant attention in recent years due to its capability to analyze medical data and support disease prediction. Diabetes mellitus, being one of the most prevalent chronic diseases worldwide, has become an important area of research in predictive healthcare analytics. Early identification of individuals at risk can support timely medical intervention and preventive healthcare management.

Several research studies have been carried out to predict diabetes using patient clinical and demographic attributes. These studies employ various machine learning and data analytics techniques to improve prediction performance and assist in medical decision-making. However, many of the existing approaches focus primarily on prediction accuracy and provide limited interpretability and practical usability. This chapter presents a detailed review of existing diabetes prediction systems, their limitations, and the need for the proposed explainable and user-friendly framework developed in this project.

2.1 EXISTING SYSTEM

Many existing diabetes prediction systems are based on supervised machine learning algorithms that classify patients as diabetic or non-diabetic based on health-related input features. Commonly used algorithms include Logistic Regression, Decision Tree, Support Vector Machine, Random Forest, K-Nearest Neighbors, Naïve Bayes, and XGBoost**.** These models are typically trained using structured healthcare datasets containing attributes such as age, BMI, glucose level, HbA1c level, hypertension, heart disease, gender, and smoking history.

A significant recent contribution in this field is the study by Barman et al. [1], which applied interpretable machine learning techniques for diabetes risk prediction using large-scale Indian national survey data. The study demonstrated that machine learning can effectively identify diabetes risk patterns across diverse populations while emphasizing the importance of interpretability in healthcare analytics. Their

work highlights that accurate prediction alone is not sufficient unless the model's decisions can also be explained clearly.

Similarly, Netayawijit et al. [2] proposed an interpretable machine learning framework for diabetes prediction by integrating SMOTE balancing with SHAP-based explainability. Their work focused not only on improving predictive performance but also on enhancing transparency for clinical decision support. This study is especially relevant because it combines class balancing and explainability, both of which are important in healthcare prediction systems.

Another important study by Barzegar et al. [3] introduced a machine learning pipeline for early diabetes detection and performed a comparative analysis of multiple models along with explainable AI techniques. Their findings suggest that advanced ensemble methods can provide better predictive performance while explainability tools improve the understanding of model behavior.

The study titled "A novel machine learning approach for diagnosing diabetes with a self-explainable interface" [4] further emphasized the need for systems that are not only accurate but also understandable to users. This work attempted to bridge the gap between machine learning prediction and user interpretation by introducing an interface that supports self-explanation.

Likewise, the study "Towards Transparent Diabetes Prediction: Combining AutoML and Explainable AI for Improved Clinical Insights" [5] focused on improving diabetes prediction by combining automated machine learning with explainable AI methods. The research demonstrated that interpretability and transparency are essential for making intelligent healthcare systems more understandable, trustworthy, and clinically useful.

In addition to these application-level studies, foundational work by Lundberg and Lee [6] introduced SHAP (SHapley Additive Explanations), which has become one of the most widely used methods for explaining machine learning model predictions. SHAP enables feature-level interpretation by quantifying how much each input contributes to the final prediction. This method has become especially valuable in healthcare applications where model transparency is critical.

On the modeling side, Chen and Guestrin [7] introduced XGBoost, a scalable tree boosting algorithm that has shown excellent performance on structured tabular datasets. XGBoost is widely used in medical prediction tasks due to its high accuracy, ability to handle feature interactions, and robustness against overfitting.

Furthermore, publicly available healthcare datasets such as the Diabetes Prediction Dataset by Tusher [8] have enabled researchers to build and evaluate machine learning models efficiently. Such datasets contain relevant patient information and are widely used for benchmarking diabetes prediction systems.

Although these existing systems and studies have made valuable contributions to diabetes risk prediction, many of them remain limited to research environments and do not fully integrate prediction, explainability, user interaction, visualization, and personalized recommendation in a single practical application.

2.2 DISADVANTAGES OF EXISTING SYSTEM

Despite the progress in diabetes prediction research, existing systems still have several limitations that reduce their practical usefulness. One of the major drawbacks is the black-box nature of many machine learning models. Although advanced algorithms provide good accuracy, they often fail to clearly explain how the prediction is generated, which reduces trust in healthcare applications [1][5].

Another limitation is that many systems focus mainly on performance metrics such as accuracy, precision, recall, and F1-score, while giving less importance to explainability**.** Without proper interpretation, users cannot easily understand which health factors, such **as** HbA1c level, blood glucose, BMI, or smoking history, contribute most to diabetes risk [2][3][6].

In addition, many existing systems are limited to research or notebook-based implementations and do not provide a user-friendly interface for practical use. Most of them also produce only a binary output such as diabetic or non-diabetic, without categorizing the result into meaningful risk levels like LOW, MEDIUM, and HIGH.

Furthermore, many systems do not provide personalized recommendations, visual explanations, or a clinical-style summary, which reduces their value as a real-

world healthcare support tool. Therefore, existing systems are often limited in terms of interpretability, usability, and practical deployment.

2.3 PROPOSED SYSTEM

To overcome the limitations of existing systems, the proposed project introduces an Explainable Diabetes Risk Prediction System that combines machine learning, explainability, visualization, and personalized healthcare support in a single platform.

The system is developed as a Streamlit-based web application where users can enter important health-related parameters such as age, BMI, HbA1c level, blood glucose level, hypertension, heart disease, gender, and smoking history. Based on these inputs, the trained machine learning model predicts the probability of diabetes risk and classifies the result into meaningful categories such as LOW, MEDIUM, and HIGH risk for better understanding and interpretation.

To improve transparency, the proposed system integrates SHAP (SHapley Additive Explanations), which helps explain how each feature contributes to the final prediction. This allows users to clearly understand the major factors influencing diabetes risk, making the model more interpretable and trustworthy in a healthcare context [2][6].

In addition to prediction, the system also provides feature importance analysis, patient-specific explanation, clinical summary, and personalized recommendations based on the user's health condition. These recommendations include guidance related to diet, lifestyle changes, weight management, blood sugar monitoring, and medical consultation when necessary.

The proposed system is designed not only to predict diabetes risk but also to support early awareness and preventive healthcare decision-making. By combining prediction with explanation and health guidance, the system helps users better understand their current health condition and encourages them to take timely preventive actions. This makes the application more practical and useful compared to traditional prediction-only systems.

Thus, the proposed system is designed to be more accurate, transparent, user-friendly, and practically useful for early diabetes risk assessment and preventive healthcare support.

2.4 ADVANTAGES OF PROPOSED SYSTEM

The proposed Diabetes Risk Prediction System offers several important advantages over traditional and existing diabetes prediction methods. It is designed not only to provide accurate prediction results but also to improve interpretability, usability, and practical healthcare support. By combining machine learning, explainable artificial intelligence, and an interactive web interface, the application becomes more useful for both academic and healthcare-related purposes.

One of the major advantages of this system is its ability to support early diabetes risk assessment in a simple and efficient manner. The application analyzes important health-related parameters such as age, BMI, HbA1c level, blood glucose level, hypertension, heart disease, gender, and smoking history to estimate the probability of diabetes. This helps in identifying individuals who may require preventive care or medical attention at an early stage.

Another important advantage is the classification of results into meaningful risk levels such as LOW, MEDIUM, and HIGH. Unlike many conventional systems that provide only a diabetic or non-diabetic output, this approach improves understanding by presenting the prediction in a more informative way. This makes the result easier to interpret and more useful for healthcare-related decision-making.

A key strength of the application is the integration of Explainable Artificial Intelligence (XAI) using SHAP (SHapley Additive Explanations). This helps explain how each feature contributes to the final prediction. As a result, users can understand whether factors such as high HbA1c level, elevated blood glucose, obesity, smoking history, or hypertension are increasing their diabetes risk. This improves the transparency, trust, and interpretability of the application.

The system also provides patient-specific clinical interpretation, which is another major advantage. Instead of displaying only raw prediction scores, the

application identifies the most influential factors affecting the current user's result and presents them in a more understandable form. This helps users gain better awareness of their individual health condition and the possible medical significance of their input values.

Another useful feature is the inclusion of personalized health recommendations. Based on the prediction result and user-provided health parameters, the application generates customized suggestions related to blood sugar monitoring, weight management, healthy diet, lifestyle improvement, and medical consultation when necessary. This makes the system more practical and useful than a simple prediction-only model.

The application further improves healthcare support by providing food and diet recommendations based on the user's risk category. Users identified as HIGH risk receive stricter diabetic diet guidance, while MEDIUM risk users are advised to follow controlled dietary habits and LOW risk users are encouraged to maintain a healthy lifestyle. This feature adds real-world value to the system and supports preventive healthcare awareness.

Another significant advantage is its interactive and user-friendly web interface developed using Streamlit. The application allows users to enter their health details easily and obtain prediction results instantly. It also includes visual analytics and graphical interpretation, such as feature importance charts and SHAP-based explanation plots, which make the results easier to understand and improve the analytical value of the application.

Finally, the system combines prediction, explanation, visualization, clinical summary, and personalized recommendations into a single integrated platform. This makes it more accurate, transparent, user-friendly, and practically useful than many existing diabetes prediction systems. Therefore, the application serves not only as a prediction tool but also as a preventive healthcare support and awareness system.

3 ANALYSIS

The analysis phase is an important part of the software development process in which the requirements, structure, and feasibility of the system are studied in detail. It helps in identifying the user needs, system functionalities, and technical resources required for the successful development of the application. A proper analysis ensures that the system is designed efficiently and can satisfy both functional and practical requirements.

In this project, the analysis focuses on the development of Diabetes Prediction System, which combines machine learning, data analytics, explainable artificial intelligence, and an interactive web-based interface. The system is designed to provide accurate diabetes prediction, clear explanation of results, personalized recommendations, and downloadable report generation. Therefore, this chapter presents the requirement specification and the overall content flow of the proposed project.

3.1 SOFTWARE REQUIREMENT SPECIFICATION

The Software Requirement Specification (SRS) defines the complete requirements of the proposed Diabetes Prediction Using Data Analytics and Machine Learning system. It describes the user needs, software tools, and hardware environment required for smooth operation. The main objective of this system is to provide an accurate, interpretable, and user-friendly platform for predicting diabetes risk using machine learning techniques.

The system processes user-provided clinical and demographic data to generate a diabetes risk probability. The output includes a risk category, feature-based explanations, graphical visualizations, and personalized health recommendations, ensuring that the system is reliable, responsive, and suitable for real-time use.

3.1.1 User Requirements

The system is designed to be intuitive, interactive, and accessible for all users, including those without technical expertise. Users are required to submit their personal and clinical information relevant to diabetes assessment, such as general health indicators, lifestyle factors, and medical history. Once the data is entered,

the system processes it and provides an immediate prediction of the user's diabetes risk.

The prediction is presented clearly, including a risk category, feature-based explanations, graphical visualizations, and personalized health recommendations. The system ensures that users can understand how various factors contribute to their risk and take informed preventive actions. It must provide a responsive and user-friendly interface, maintain high accuracy and reliability, and be suitable for real-time use in practical healthcare scenarios.

3.1.2 Software Requirements

The software requirements define the essential tools, libraries, and frameworks needed for the development and deployment of the Diabetes Prediction System. They ensure that the system can process clinical data efficiently, execute machine learning algorithms accurately, generate visual explanations, and provide an interactive interface for real-time user interaction. Properly defining these requirements contributes to reliability, scalability, and usability of the system, ensuring that predictions are accurate, explanations are interpretable, and the system is practical for healthcare users.

Table 3.1 Software Requirements

S. No	Requirement Type	Description
1	Programming Language	Python 3.8+ for flexibility and support of machine learning libraries
2	Development Tools	VS Code, Jupyter Notebook, Google Colab for coding, testing, and prototyping
3	Data Processing	Pandas, NumPy for handling and preprocessing data
4	Machine Learning	Scikit-learn, XGBoost for training and evaluating predictive models
5	Explainability	SHAP for feature contribution analysis and transparent predictions

6	Visualization	Matplotlib, Plotly for charts, graphs, and SHAP visualizations
7	Web Framework	Streamlit for building interactive, user-friendly web interface
8	Additional Tools	Joblib for model saving/loading and caching artifacts

The listed software requirements collectively enable the Diabetes Prediction System to function accurately, efficiently, and interactively. Python provides the programming environment, while libraries like Pandas and NumPy ensure smooth handling, cleaning, and preprocessing of clinical and demographic data. These tools are essential for preparing the data in a format suitable for machine learning analysis.

Machine learning libraries such as Scikit-learn and XGBoost allow the system to analyze the data and generate reliable diabetes risk predictions. SHAP enhances interpretability by explaining the contribution of each feature to the prediction, while visualization libraries like Matplotlib and Plotly present the results in a clear, graphical format. The Streamlit framework ensures that users can interact with the system, submit data, and receive predictions in real time through an intuitive web interface.

Additional tools such as Joblib facilitate saving and loading trained models efficiently, ensuring fast and consistent predictions without retraining. Together, these software requirements provide a robust foundation for the project, allowing it to deliver risk probabilities, categorized risk levels, graphical explanations, and personalized health recommendations. They ensure that the system is reliable, interpretable, responsive, and practical for real-time diabetes risk assessment.

3.1.3 Hardware Requirements

The hardware requirements define the minimum system configuration needed to ensure smooth execution of the Diabetes Prediction System. Proper hardware selection is essential to handle data processing, model computations, and visualization efficiently. The system is designed to be compatible with commonly available hardware to maximize accessibility.

Table 3.2 Hardware Requirements

Component	Specification	Contribution to Project
Operating System	Windows10/Linux/ macOS	Provides a stable environment to run Python and Streamlit application, ensuring compatibility across platforms.
Processor (CPU)	Intel i3 or equivalent	Handles data preprocessing, feature scaling, and execution of trained machine learning models efficiently.
RAM	Minimum 4 GB	Ensures smooth processing of input data, SHAP-based explanations, and interactive visualizations in real-time.
Hard Disk	Minimum 250 GB	Stores datasets, trained model files, and necessary libraries to support system operation and scalability.

The hardware requirements table outlines the essential components needed to run the Diabetes Prediction System effectively. The operating system serves as the foundation for executing the Python-based application and the Streamlit interface. Compatibility with Windows, Linux, and macOS ensures that the system can be accessed by a wide range of users across multiple platforms without any restrictions.

The processor and RAM specifications are critical for handling computational tasks efficiently. A minimum Intel i3 processor provides sufficient processing power for data preprocessing, feature scaling, and execution of the trained machine learning models. Likewise, at least 4 GB of RAM ensures smooth handling of input data, real-time predictions, and generation of SHAP-based explanations and visualizations, preventing performance delays during interactive use.

Adequate storage capacity is also necessary to maintain the system's functionality and efficiency. A minimum of 250 GB hard disk space is recommended to store datasets, trained model files, and supporting libraries.

This ensures that the application operates smoothly and can scale to handle larger datasets or additional computational tasks in the future. Together, these hardware components provide a reliable foundation for a responsive and user-friendly diabetes prediction application.

3.2 CONTENT DIAGRAM OF PROJECT

The workflow diagram illustrates the step-by-step process of the Diabetes Prediction System, highlighting how user input flows through various modules to produce interpretable predictions and personalized recommendations.

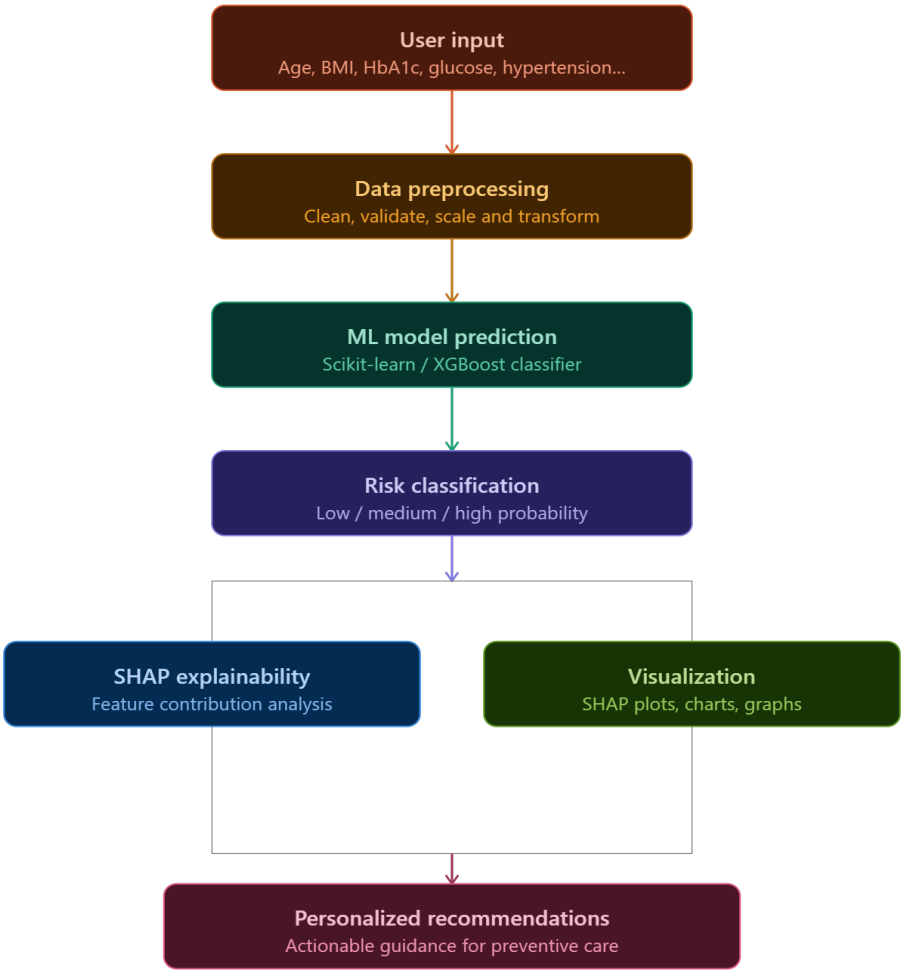

Fig 3.1 Content Diagram

- User Input: The process begins with the user entering health-related parameters, including age, BMI, HbA1c level, blood glucose, hypertension, heart disease, gender, and smoking history. These inputs form the basis for evaluating the probability of diabetes.
- Data Preprocessing: The collected data is validated, cleaned, scaled, and transformed to ensure consistency and compatibility with the machine learning model. Proper preprocessing is crucial for accurate predictions and avoids errors due to out-of-range or inconsistent values.
- ML Model Prediction: The preprocessed data is fed into the trained machine learning model, which can include algorithms like Scikit-learn classifiers or XGBoost. The model calculates the likelihood of the user developing diabetes based on historical patterns learned from the training dataset.
- Risk Classification: The predicted probability is categorized into low, medium, or high-risk levels, making the results more interpretable for users and helping them understand their relative risk at a glance.
- SHAP Explainability: The system computes SHAP (SHapley Additive Explanations) values to identify how each input feature contributes to the prediction. This provides transparency and allows users to understand which health factors most influence their diabetes risk.
- Visualization: The system presents the SHAP results and other insights using intuitive charts, plots, and graphs. These visualizations make complex model outputs easier to interpret and support informed decision-making.
- Personalized Recommendations: Based on the identified risk factors and overall risk classification, the system generates actionable lifestyle and preventive care suggestions. This ensures users receive guidance tailored to their health profile, supporting preventive interventions and better management of diabetes risk

4 DESIGN

The design phase of the Diabetes Prediction System defines the overall architecture and structure of the application, detailing how different modules interact to achieve accurate predictions, interpretability, and user-friendly recommendations. This phase focuses on creating a modular, scalable, and maintainable system that integrates data preprocessing, machine learning prediction, SHAP-based explanations, visualizations, and personalized recommendations. Clear design ensures that the system operates efficiently while providing transparency and actionable insights to the user.

The design emphasizes **modularity**, allowing each component to operate independently while communicating seamlessly with other modules. This ensures that future enhancements, such as adding more features or upgrading the prediction model, can be implemented without affecting the overall system functionality.

4.1 INTRODUCTION

The design phase focuses on translating system requirements into a structured blueprint that guides development. It ensures that the system is efficient, scalable, maintainable, and user-friendly while providing a clear understanding of how different components interact and function. This phase defines the overall architecture, module interactions, and the flow of data from input to output.

The design phase typically includes several key components: Data Flow Diagrams (DFD) to illustrate the movement of data within the system; UML diagrams**,** including Use Case, Sequence, and Class Diagrams, to represent user interactions, system behavior, and structural organization; Entity-Relationship (ER) Diagrams to model database structures when applicable; and a Data Dictionary to define all input and output data elements, their types, allowed values, and descriptions. These components collectively provide a comprehensive understanding of system functionality, structure, and data management.

Additionally, the design phase establishes the interactions between various modules, ensuring seamless communication and integration. By clearly defining module responsibilities, data flow, and system architecture, the design phase lays the foundation for a smooth and efficient implementation.

4.2 DFD/ ER/ UML DIAGRAM

This section describes the various design diagrams that visually represent the system's structure, behavior, and data flow. These diagrams provide a clear understanding of how the system operates, how data is processed, and how different modules interact.

Data Flow Diagram (DFD): Illustrates how data moves within the system, showing processes, data stores, and external entities. It provides a high-level view of how inputs are transformed into outputs.

Entity-Relationship (ER) Diagram: Represents the data structure, showing entities, their attributes, and the relationships between them. This diagram helps in understanding how the system organizes and manages data efficiently.

UML Diagrams: Capture both structural and behavioral aspects of the system:
Use Case Diagram: Shows user interactions with the system and the available functionalities.
Sequence Diagram: Details the step-by-step interactions between users and system components during various processes.
Class Diagram: Illustrates the structural organization, showing classes, their attributes, methods, and relationships among them.

Together, these diagrams provide a comprehensive visualization of the system design. They clarify the workflow, module interactions, data relationships, and overall system architecture, forming a foundation for efficient implementation.

4.2.1 Data Flow Diagram (DFD)

At the context level (Level 0), the primary external entity is the User/Patient, who provides essential medical and personal data such as age, BMI, HbA1c levels, glucose levels, and other relevant health indicators. This information is fed into the system, which processes it to generate meaningful outputs including the predicted diabetes risk, risk percentage, categorized risk level (low, medium, or high), interpretability insights using SHAP, visualizations, and personalized health recommendations. These results are then communicated back to the user/patient.

At DFD Level 1, the process begins with Collect Patient Input (1.0), where all necessary health details are gathered from the user. This is followed by Data Preprocessing (2.0), which ensures the quality and usability of the data by validating inputs, encoding categorical variables, and scaling numerical values. The processed data is stored in the Patient Input Data (D1) repository, which serves as a reference for subsequent steps.

Next, in ML Model Prediction (3.0), the system utilizes a pre-trained machine learning model stored in the Trained ML Model (D2) data store to generate predictions based on the processed input data. The output from this stage is then passed to Risk Classification (4.0), where the system categorizes the predicted risk into low, medium, or high levels for easier interpretation.

Following this, the SHAP Explanation & Visualization (5.0) stage provides transparency by explaining how different features contributed to the prediction. It generates visual insights that help users understand the reasoning behind the model's output. Finally, in Recommendation Generation (6.0), the system produces personalized suggestions aimed at reducing diabetes risk, such as lifestyle changes or medical advice. These final results are delivered back to the user/patient, completing the data flow cycle

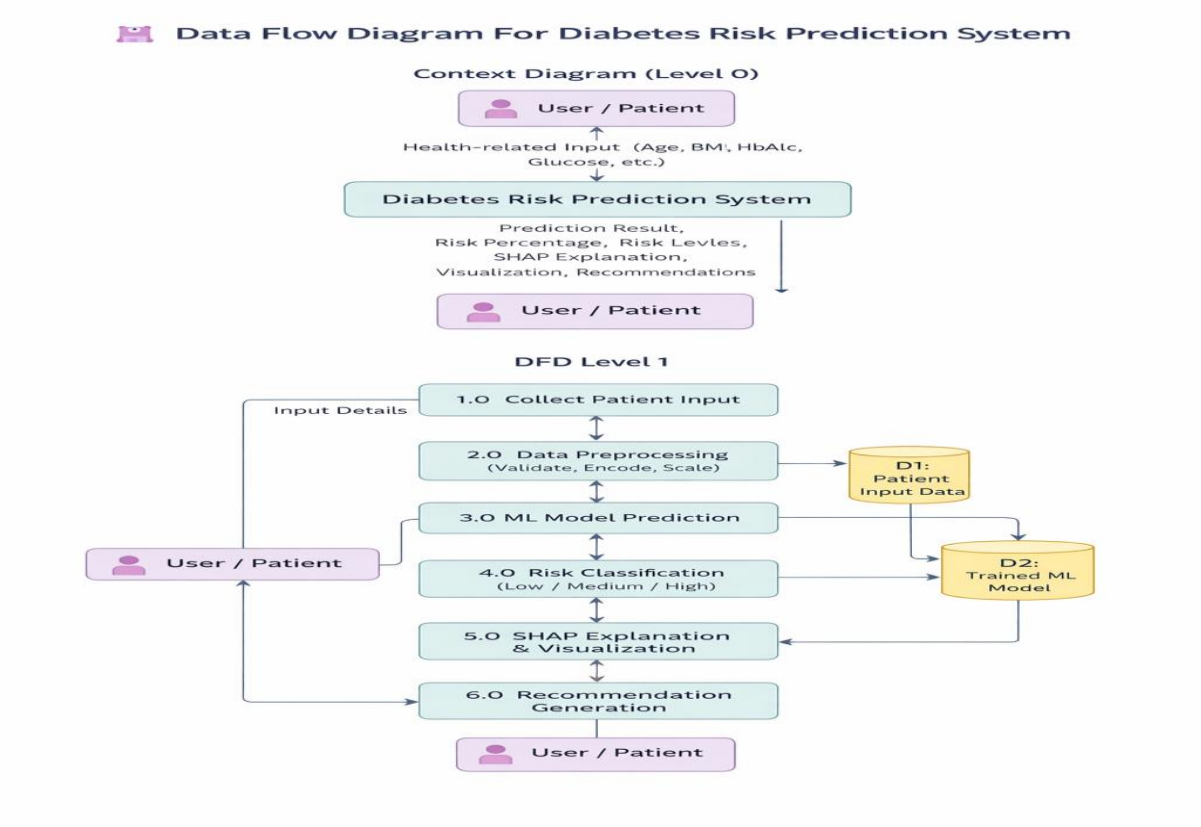

Fig 4.1 Data Flow Diagram

4.2.2 Entity-Relationship (ER) Diagram

The diagram represents a simple database structure for a Diabetes Risk Prediction System, consists of multiple interconnected entities that collectively manage patient data, predictions, model training, and decision support outputs. Each entity plays a specific role in ensuring accurate prediction, interpretability, and recommendation generation.

The Patient entity represents the user of the system who provides health-related information. It typically contains attributes such as patient ID, name, age, gender, BMI, and other relevant medical details. This entity acts as the central point of interaction, as all predictions and recommendations are associated with a specific patient. A patient can submit multiple records over time, allowing the system to track historical health trends and predictions.

The Prediction entity stores the results generated by the system after processing patient input. It includes attributes such as prediction_id (primary key), patient_id (foreign key), predicted risk value, risk percentage, and

timestamp of prediction. This entity captures the outcome of the machine learning model and maintains a history of predictions for each patient. The relationship between Patient and Prediction is typically one-to-many, as a single patient can have multiple prediction records.

The Training_Dataset entity contains the historical data used to train the machine learning model. It includes attributes such as record_id (primary key), HbA1c level, glucose level, BMI, age, and other clinical features relevant to diabetes prediction. This dataset is crucial for building the predictive model, as it allows the system to learn patterns and correlations between input features and diabetes outcomes. It is not directly linked to individual users but serves as the backbone for model accuracy and performance.

The ML Model (Trained Model) entity represents the trained machine learning model used for generating predictions. It may include attributes such as model ID, model type (e.g., logistic regression, random forest), training accuracy, and version. This entity is connected to both the training dataset (used to build the model) and the prediction process (used to generate outputs). Storing model details helps in version control, performance tracking, and future improvements.

The Risk Classification component categorizes the raw prediction output into meaningful groups such as low, medium, or high risk. Although it may not always be stored as a separate table, it is an important logical entity in the system. It simplifies interpretation for users and enables easier decision-making based on predefined thresholds.

The SHAP Explanation entity is responsible for providing interpretability to the model's predictions. It contains feature importance values that explain how each input factor contributed to the final prediction. This enhances transparency and trust in the system, especially in healthcare applications where understanding the reasoning behind predictions is critical.

The Visualization entity works alongside SHAP explanations to present data in graphical formats such as charts or plots. It helps users better understand their health status and the factors influencing their risk level. Visual outputs improve user engagement and make complex model results easier to interpret.

Finally, the Recommendation entity generates personalized suggestions based on the patient's risk level and input data. These recommendations may include lifestyle changes, dietary advice, exercise routines, or suggestions to consult a healthcare professional. This entity ensures that the system goes beyond prediction by providing actionable insights to improve patient health outcomes.

Overall, these entities are interconnected to form a comprehensive system where patient data is collected, processed, analyzed using machine learning, and transformed into meaningful predictions, explanations, and recommendations. This structured design ensures efficient data management, accurate predictions, and user-friendly outputs in the diabetes risk prediction system.

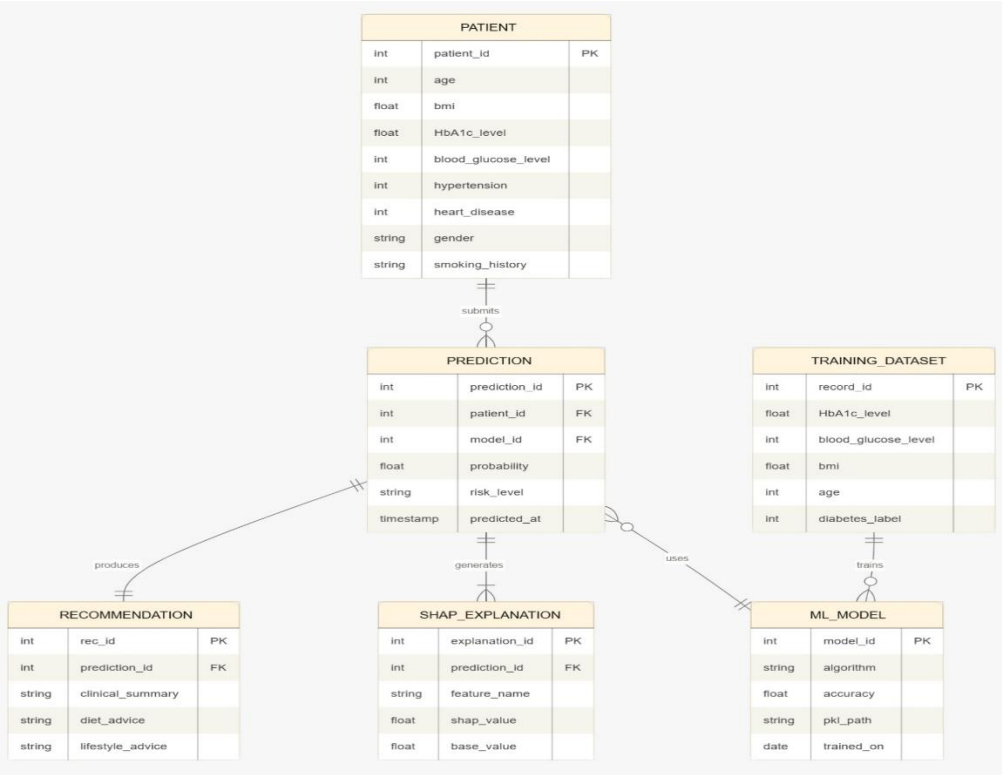

Fig 4.2 Entity-Relationship Diagram

4.2.3 Use Case Diagram

The Diagram illustrates how the primary actor, the Patient, interacts with the system to assess their diabetes risk and receive meaningful insights. The patient initiates the process by submitting health data, which includes medical parameters such as age, BMI, glucose levels, and HbA1c. This interaction

represents the entry point into the system and triggers a sequence of internal processes.

Once the data is submitted, the system performs the collect patient data and preprocess data use cases. During preprocessing, the system validates, encodes, and scales the input data to ensure it is suitable for analysis. After preprocessing, the system proceeds to the generate risk prediction use case, where the machine learning model analyzes the input data to estimate the likelihood of diabetes.

The prediction process is closely associated with the risk classification use case, which categorizes the predicted outcome into different levels such as low, medium, or high risk. This classification helps simplify complex prediction results into understandable categories for the patient. The ML Model acts as a supporting component in this stage, providing the trained logic required for accurate prediction and classification.

To enhance transparency, the system includes the explain prediction with SHAP use case. This allows the patient to understand how different input features contributed to the prediction. Alongside this, the system also performs generate visualization, presenting the results in graphical formats that improve clarity and user comprehension.

Finally, the system executes the generate recommendations use case, where personalized advice is provided based on the patient's risk level. These recommendations may include lifestyle changes, dietary suggestions, or medical consultations. The patient can then receive prediction results, view SHAP explanations, and receive recommendations, completing the interaction cycle.

Overall, the use case diagram demonstrates a user-centered workflow where the patient interacts with the system to input data and receive predictions, explanations, visualizations, and actionable health recommendations, ensuring both usability and interpretability.

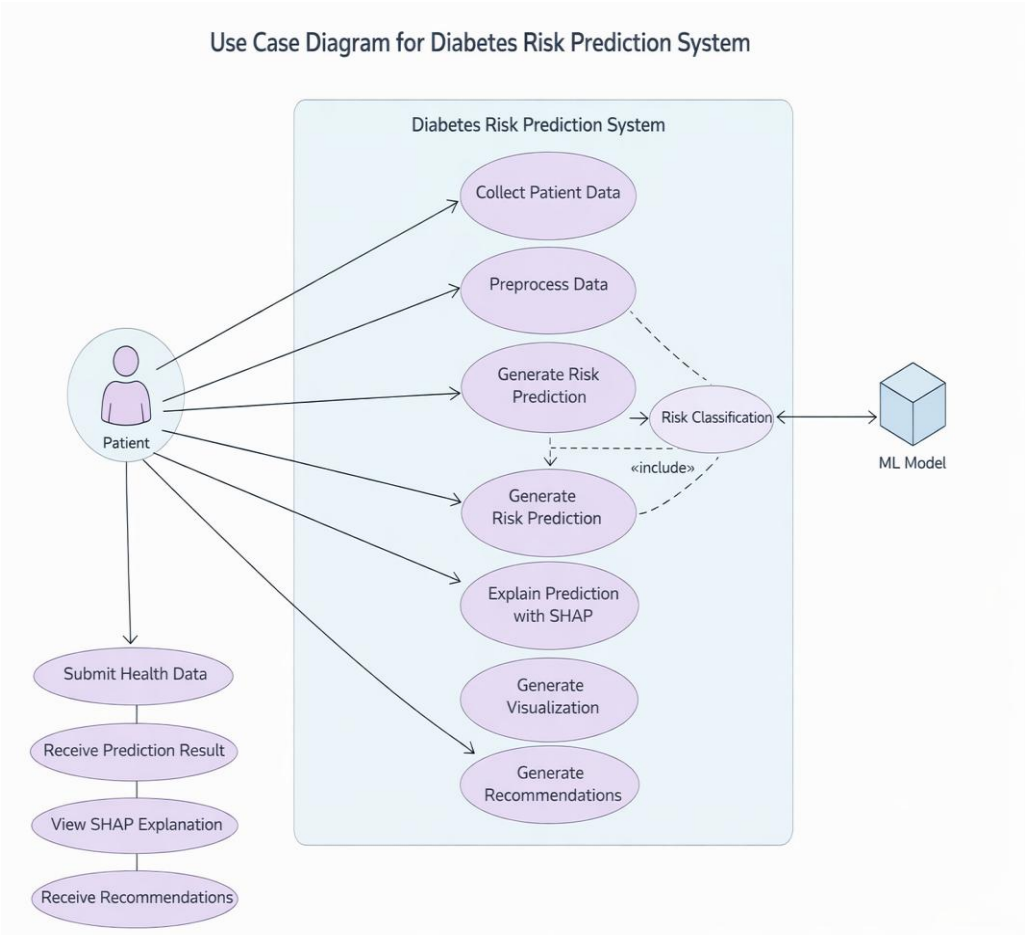

Fig 4.3 Use Case Diagram

4.2.4 Sequence Diagram

The Diagram illustrates the step-by-step interaction between different system components involved in processing user input and generating predictions. The main participants in this sequence are the User, StreamlitApp, PatientInput, DiabetesPredictor, SHAPExplainer, and ReportGenerator, each playing a specific role in the workflow.

The process begins when the User submits health data through the interface. This request is received by the StreamlitApp, which acts as the central controller of the application. The StreamlitApp then invokes the PatientInput component to collect and structure the input data, converting it into a suitable format using methods like to_dataframe().

Once the input data is prepared, it is sent for preprocessing and passed to the DiabetesPredictor. The predictor scales the input data and uses the

trained machine learning model to generate a prediction, producing both the risk probability and risk level (such as low, medium, or high). These prediction results are then returned to the StreamlitApp.

Next, the processed and scaled data is forwarded to the SHAPExplainer, which computes SHAP values to explain the prediction. This step identifies the contribution of each feature to the final result and helps improve transparency. The SHAPExplainer may also rank features and prepare data for visualization.Following the explanation phase, the system calls the ReportGenerator, which uses the prediction results and feature importance data to generate personalized recommendations and a summary report. This includes health advice, risk interpretation, and possibly visual indicators such as risk badges.

Finally, the StreamlitApp displays the results to the user, including the prediction outcome, SHAP-based explanations, visualizations, and recommendations. If the input data is invalid or incomplete, an alternative flow is triggered where the system displays an error message instead of proceeding with prediction.

Overall, the sequence diagram clearly shows the logical flow of operations, highlighting how user input moves through different components of the system to produce accurate, interpretable, and actionable outputs in a structured and sequential manner.

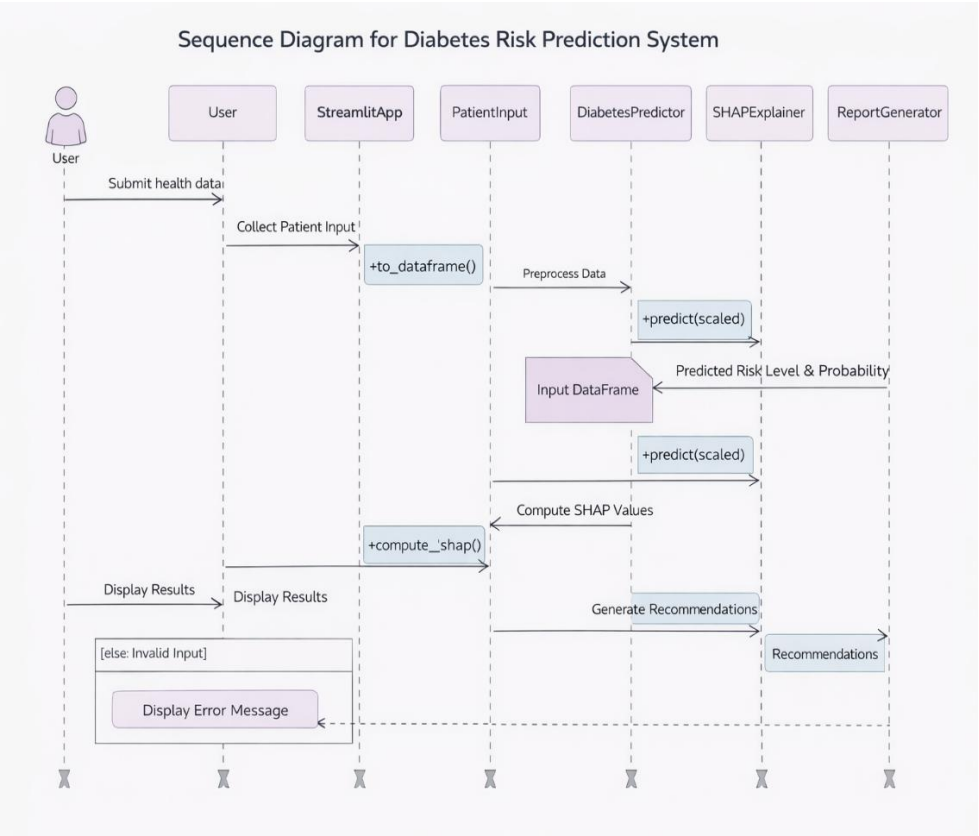

Fig 4.4 Sequence Diagram

4.2.5 Class Diagram

The diagram represents the architecture of the Diabetes Risk Prediction System implemented using the XGBoost model, highlighting the interaction between various system components. The system is composed of key classes including StreamlitApp, PatientInput, ModelTrainer, DiabetesPredictor, SHAPExplainer, and ReportGenerator, each contributing to a specific stage in the workflow from data input to final output.

The StreamlitApp class serves as the main interface and controller of the application. It maintains the session state and provides methods to render different sections of the user interface, such as input collection, prediction results, SHAP explanations, and reports. It captures user input through the sidebar, triggers prediction when the user clicks a button, and coordinates communication with backend components like the predictor, explainer, and report generator.

The PatientInput class is responsible for collecting and structuring patient health data. It includes attributes such as age, BMI, HbA1c level, blood glucose level, hypertension status, heart disease, gender, and smoking history. The class provides methods like to_dataframe() and encode_features() to convert raw user input into a structured and machine-readable format suitable for the model.

The ModelTrainer class is used to build and manage the machine learning model using XGBoost (XGBClassifier). It includes attributes such as dataset path and the model itself. Its methods handle loading the dataset, preprocessing data, training the model, evaluating performance, and saving trained artifacts. This component ensures that the model is properly trained and optimized before being deployed for predictions.

The DiabetesPredictor class is the core prediction engine that uses the trained XGBoost model. It contains the model, a scaler for feature normalization, and a list of feature names. Its methods allow loading saved model artifacts, scaling input data, generating prediction probabilities, and classifying the risk level. Compared to traditional models, XGBoost enhances prediction accuracy and performance through gradient boosting techniques.

The SHAPExplainer class provides interpretability for the XGBoost model predictions. Using a TreeExplainer, it computes SHAP values to determine the contribution of each feature to the prediction. It also ranks features and generates visualizations, helping users understand the reasoning behind the predicted risk level.

Finally, the ReportGenerator class is responsible for producing user-friendly outputs. It generates AI-based summaries, personalized health recommendations, and visual indicators such as risk badges. It uses inputs like prediction probability, risk level, feature importance, and patient data to create meaningful and actionable insights.

Overall, this class diagram demonstrates a modular and scalable design where StreamlitApp manages user interaction, PatientInput handles data collection, ModelTrainer builds the XGBoost model, DiabetesPredictor

performs predictions, SHAPExplainer ensures transparency, and ReportGenerator delivers final recommendations. The use of XGBoost improves prediction accuracy while maintaining interpretability through SHAP, making the system both powerful and user-friendly.

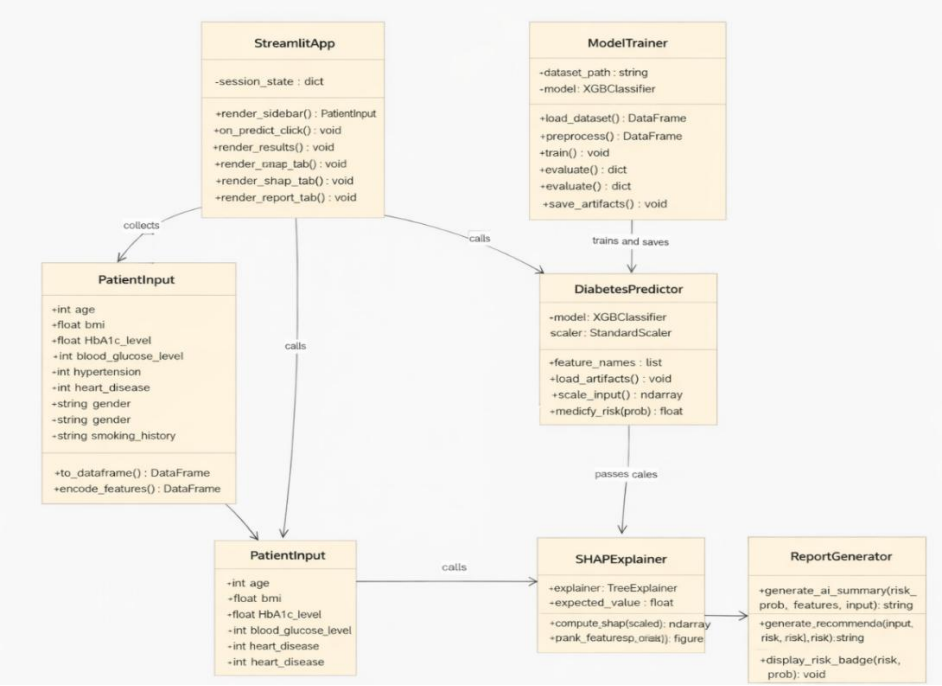

Fig 4.5 Class Diagram

4.3 DATA DICTIONARY

A data dictionary provides a detailed description of the data elements used in the system. It helps in understanding the nature of the input and output values, their types, and their purpose within the application. This ensures consistency in data handling, improves system clarity, and supports smooth implementation and maintenance.

4.3.1 Input Data Dictionary

The input data dictionary defines the health-related attributes entered by the user. These values are used by the machine learning model to analyze patterns and predict the likelihood of diabetes.

Table 4.1 Input Data Dictionary

Field Name	Data Type	Description
Gender	Categorical	Represents the gender of the user
Age	Integer	Indicates the age of the user in years
Hypertension	Binary / Integer	Shows whether the user has hypertension
Heart Disease	Binary/Integer	Indicates whether the user has heart disease
Smoking History	Categorical	Represents the smoking habit of the user
BMI	Float	Body Mass Index of the user
HbA1c Level	Float	Represents the average blood sugar level over the past 2–3 months
BloodGlucose Level	Integer	Indicates the current blood glucose measurement of the user

4.3.2 Output Data Dictionary

The output data dictionary defines the values generated by the system after processing the user input through the machine learning model and explainability modules.

Table 4.2 Output Data Dictionary

Field Name	Data Type	Description
Prediction Probability	Float	Diabetes risk score
Risk Level	Categorical	Risk category
Prediction Result	Text	Final prediction
SHAP Explanation	Text / Numerical	Feature contribution
Visualization Output	Graphical	Graphical result display
Personalized Recommendations	Text	Suggested health advice

5 IMPLEMENTATION & RESULTS

This section describes how the designed system is developed into a working application and how its functionalities are executed in practice. It explains the implementation of various modules, tools, and algorithms used in the system. It also presents the outputs and results obtained from the application, demonstrating its performance and effectiveness.

5.1 METHOD OF IMPLEMENTATION

The implementation of the system is carried out using Python, Streamlit, Machine Learning algorithms, Scikit-learn, SHAP, and visualization libraries. These technologies are used to build an interactive application that can process user input, generate predictions, explain feature contributions, and display the results in a clear and understandable manner.

The implementation is divided into two major phases: the Model Development Phase and the Prediction Phase. These two phases ensure that the system is capable of learning from the dataset and providing accurate predictions based on user input. The first phase focuses on preparing the data and training the machine learning model, while the second phase deals with real-time user interaction and prediction generation.

Model Development Phase: The model development phase focuses on building a reliable machine learning model capable of predicting diabetes risk using historical health-related data. For this purpose, the Diabetes Prediction Dataset [8] is used, which contains important input features such as gender, age, hypertension, heart disease, smoking history, BMI, HbA1c level, and blood glucose level. These attributes serve as the basis for identifying patterns associated with diabetes.

In this phase, the dataset is first loaded and examined to understand its structure, feature types, and class distribution. Data preprocessing techniques are then applied to prepare the dataset for model training. This includes separating categorical and numerical features, applying appropriate

transformations, and converting the dataset into a suitable format for machine learning algorithms. The processed data is then divided into training and testing sets to evaluate model performance effectively.

Multiple machine learning algorithms are trained and tested during this phase to identify the model that produces the most accurate and reliable predictions. Performance evaluation metrics such as accuracy and other classification measures are used to compare the models. Based on the evaluation results, the best-performing model is selected and saved for later use in the prediction phase of the application.

Prediction Phase: The prediction phase focuses on integrating the trained machine learning model into the application so that it can provide real-time predictions based on user input. In this phase, the user enters the required health-related details through the application interface, including age, gender, hypertension, heart disease, smoking history, BMI, HbA1c level, and blood glucose level.

Once the user submits the input, the system applies the same preprocessing steps that were used during the model development phase. The transformed input data is then passed to the saved machine learning model, which generates the prediction result and probability score. Based on the prediction probability, the system classifies the result into different risk levels such as low, medium, or high.

To improve interpretability, the prediction phase also includes SHAP-based explainability, which identifies the contribution of each input feature to the final prediction. The system further presents the results using visual outputs such as charts and plots, along with personalized health recommendations. This makes the prediction process more transparent, informative, and user-friendly.

5.1.1 Implementation of Algorithms

The implementation of algorithms involves applying suitable machine learning techniques to analyze the input data and generate diabetes prediction

results. In this phase, four supervised classification algorithms—Logistic Regression, Decision Tree, Random Forest, and XGBoost—are used to learn patterns from the dataset and identify the relationship between health-related features and diabetes risk. These models are trained, tested, and evaluated to determine the most effective algorithm, which is then integrated into the application for real-time prediction and analysis.

Logistic Regression:

Logistic Regression is a supervised machine learning algorithm commonly used for **binary classification problems**, where the output belongs to one of two possible classes. In this project, it is used to classify whether a person is likely to be **diabetic** or **non-diabetic** based on the health-related input features. Unlike linear regression, Logistic Regression uses a **sigmoid function** to convert the output into a probability value between 0 and 1, which helps in determining the final class label.

The algorithm works by analyzing the relationship between the independent variables such as age, BMI, HbA1c level, blood glucose level, hypertension, heart disease, gender, and smoking history, and the target variable indicating diabetes status. It attempts to find the best-fitting boundary that separates the two classes based on the available training data. Because of its simplicity and efficiency, Logistic Regression is often considered a good baseline model for medical classification problems.

In this project, Logistic Regression is implemented during the model development phase to evaluate how well a simple linear classification model can predict diabetes risk. It provides reasonably good performance and also helps in understanding the effect of input features on the output. The model achieved an accuracy of 88.89%, making it a useful comparative algorithm in the overall evaluation process.

Decision Tree:

Decision Tree is a supervised machine learning algorithm used for both classification and regression tasks, but in this project it is used for classification.

It works by dividing the dataset into smaller subsets based on feature values and constructing a tree-like structure of decisions. Each internal node of the tree represents a condition based on a feature, each branch represents the outcome of that condition, and each leaf node represents the final prediction.

In the context of diabetes prediction, the Decision Tree model learns how different health parameters contribute to the possibility of diabetes. For example, the model may split the data based on conditions such as high blood glucose level, elevated HbA1c level, or BMI range to determine whether the input belongs to the diabetic or non-diabetic category. This makes the model easy to interpret because the prediction is made through a sequence of understandable decision rules.

One of the major advantages of Decision Tree is that it can handle both numerical and categorical data effectively without requiring very complex preprocessing. It is also useful in identifying the most important features involved in prediction. In this project, the Decision Tree algorithm is trained and tested on the diabetes dataset and achieved an accuracy of 88.45%. Although its performance is slightly lower than some advanced models, it remains valuable because of its interpretability and simplicity.

Random Forest:

Random Forest is an ensemble machine learning algorithm that improves prediction performance by combining the outputs of multiple decision trees. Instead of relying on a single tree, Random Forest creates a collection of trees during training and produces the final prediction based on the majority vote of those trees. This approach helps reduce overfitting and increases the stability and accuracy of the model.

In this project, Random Forest is used to capture more complex relationships between health-related features and diabetes status. Each tree in the forest is trained on a random subset of the dataset and a random subset of features, which allows the model to generalize better to unseen data. This makes it particularly effective for classification tasks involving multiple medical and lifestyle factors,

such as age, hypertension, smoking history, BMI, HbA1c level, and blood glucose level.

Random Forest is advantageous because it provides better performance than a single Decision Tree and can handle large datasets with high dimensionality. It also helps in identifying feature importance, which is useful for understanding which health parameters contribute most to diabetes prediction. In this project, Random Forest achieved an accuracy of 92.80%, showing significantly better performance than Logistic Regression and Decision Tree.

XGBoost:

XGBoost, short for Extreme Gradient Boosting, is an advanced ensemble machine learning algorithm based on the concept of boosting. Unlike Random Forest, which builds trees independently, XGBoost builds trees sequentially, where each new tree is created to correct the errors made by the previous ones. This makes the algorithm highly effective in improving prediction accuracy and capturing complex patterns in structured data.

In this project, XGBoost is used to build a highly optimized diabetes prediction model using the health-related features available in the dataset. It is capable of handling both numerical and categorical patterns effectively and is known for its speed, scalability, and strong predictive performance. The algorithm works by minimizing the prediction error through iterative learning and adjusting the model to improve classification results at each step.

XGBoost is particularly suitable for medical prediction systems because it can identify subtle relationships between features that may not be easily captured by simpler models. It also performs well even when the dataset contains multiple interacting variables. In this project, XGBoost achieved the highest accuracy of 97.21%, making it the best-performing algorithm among all the implemented models. Due to its superior performance and reliability, it is selected as the final model for deployment in the application.

Among the four implemented algorithms, XGBoost achieved the highest accuracy and showed the best performance in predicting diabetes risk. While

Logistic Regression, Decision Tree, and Random Forest also produced useful results, XGBoost provided more accurate and reliable predictions. Therefore, it is selected as the final model for the system.

5.1.2 Output Screens

This section presents the important output screens generated by the application during execution. These screens help in understanding how the system accepts user input, processes the entered health parameters, and displays prediction results along with graphical analysis and explanations. The output screens demonstrate the practical implementation of the application and highlight its interactive and user-friendly design.

Fig 5.1 Home Page

The home page of the application, which serves as the entry point for the diabetes prediction system. It displays the application title, a short description, and a prediction button to guide the user toward the next step. This interface is designed to provide a simple, clear, and user-friendly start to the prediction process.

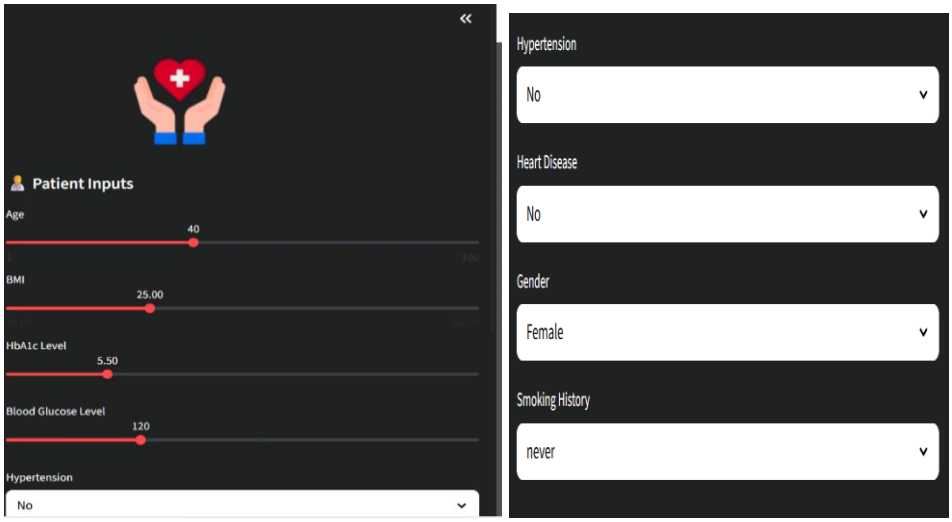

Fig 5.2 User Input Interface

The above screen represents the user input interface of the application, where the user is required to enter the essential health-related details needed for diabetes risk prediction. It includes input fields such as age, BMI, HbA1c level, blood glucose level, hypertension, heart disease, gender, and smoking history, which are considered important factors in identifying diabetes risk. This interface is designed to make data entry simple, clear, and user-friendly, ensuring that the system receives the required information accurately for further analysis and prediction.

Fig 5.3 Prediction Result Screen

After the user enters the required health-related details and submits them, the application generates the prediction output. This screen shows:

- Estimated diabetes risk percentage
- Assigned risk level such as Low, Medium, or High

The output is presented in a simple and noticeable manner so that the user can understand the prediction easily. The percentage value gives a more detailed measure of the diabetes risk, while the risk category helps in making the result easier to interpret.

This screen is important because it delivers the final prediction of the machine learning model and forms the basis for further explanation and analysis.

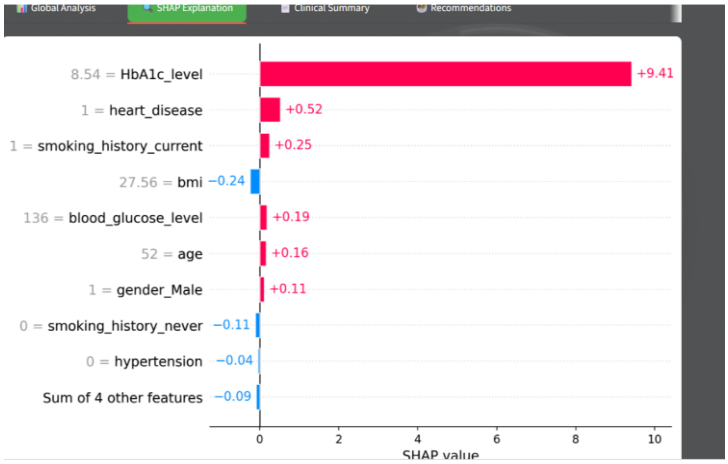

Fig 5.4 Visualization Output Screen

The SHAP Explanation screen provides a detailed interpretation of the prediction by visualising the contribution of each feature toward the final diabetes risk outcome. Features shown in pink/red increase the predicted risk (positive SHAP values) while features shown in blue reduce it (negative SHAP values).

The explanation includes:

- The most dominant risk-increasing features driving the prediction
- Secondary features that moderately contribute to or reduce the risk
- The relative importance of clinical and demographic factors
- A clear directional view of how each feature influences the model output

The chart is presented in a simple, visual format, making it easy for users without technical knowledge to understand which health factors are most responsible for their predicted risk level. This improves transparency and builds

trust in the system by making the model's decision process clearly visible and interpretable.

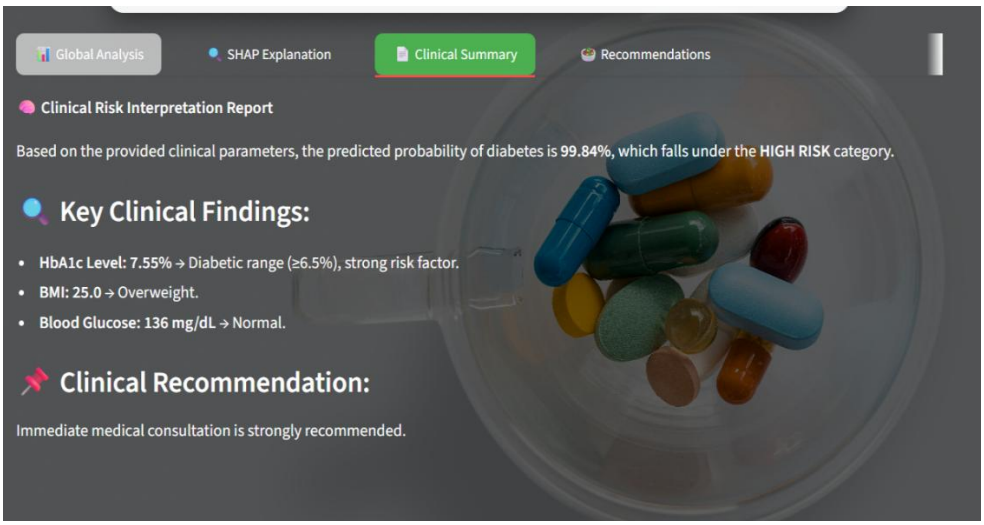

Fig 5.5 Clinical Summary Screen

The Clinical Summary screen provides a human-readable interpretation of the prediction result. It translates the model's output into clear, actionable clinical language that users without medical expertise can easily understand. The summary includes:

- The predicted probability of diabetes and its corresponding risk category
- Key clinical findings highlighting the most significant risk factors identified by the model
- A clinical interpretation of each contributing feature based on established medical thresholds
- A final clinical recommendation advising the appropriate course of action based on the overall risk level

The explanations are presented in simple, structured sentences, making it easy for users to understand the reasoning behind the prediction. This screen bridges the gap between complex machine learning outputs and practical health guidance, improving user trust and encouraging timely medical intervention.

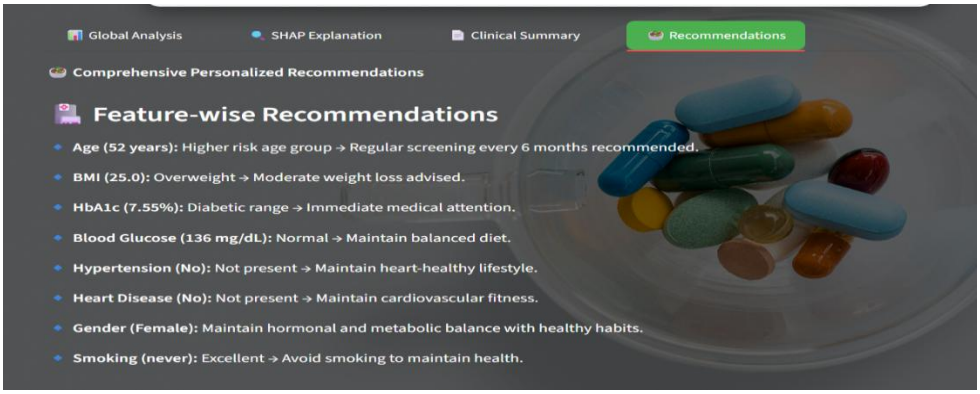

Fig 5.6 Recommendation Screen

The Recommendations screen provides comprehensive personalised health advice based on the prediction result and the user's individual feature values.

The recommendations include:

- Feature-wise guidance for each input parameter based on its clinical interpretation
- Specific actionable advice tailored to the user's age, BMI, HbA1c, blood glucose, hypertension, heart disease, gender, and smoking status
- Risk-appropriate suggestions ranging from immediate medical attention to lifestyle maintenance

This screen enhances the practical usefulness of the system by translating the prediction into real-world health actions, ensuring that every user receives relevant and personalised guidance rather than generic advice.

5.1.3 Result Analysis

The results obtained from the implemented machine learning models show that the system is capable of effectively predicting diabetes risk based on the given health-related input features. Different algorithms were trained and evaluated to compare their prediction performance, and the most suitable model was selected for deployment in the application. The analysis of the generated results confirms that the system not only produces accurate predictions but also provides meaningful interpretation through risk classification, visual analysis, and explainable outputs.

- **Model Evaluation and Selection** : XGBoost achieved the highest accuracy due to its ability to capture complex patterns and reduce errors. It was selected as the final model for its reliability, interpretability using SHAP, and robustness on structured healthcare data.
- **Accuracy Analysis** : The system achieves high prediction accuracy, with XGBoost outperforming other models. Logistic Regression provides baseline performance, while tree-based models (Random Forest, Gradient Boosting) improve predictions. Performance is evaluated using metrics such as Accuracy, Precision, Recall, and F1-Score, ensuring reliable risk classification for clinical use.
- **Prediction Output Analysis** : Predictions are expressed as probabilities and categorized into LOW, MEDIUM, and HIGH risk. Patients with elevated HbA1c, blood glucose, or high BMI are classified as high risk, while those with fewer risk factors are classified as medium or low risk.
- **Explainability and Feature Analysis** : Using SHAP, the model highlights the most influential features affecting risk. Key factors include HbA1c, blood glucose, BMI, age, and hypertension, providing transparency and building user trust.
- **Visualization and Recommendation** : Visualizations such as feature importance charts and SHAP plots make predictions easier to interpret. Personalized recommendations guide high-risk patients toward medical consultation, while medium and low-risk users receive preventive advice.

6 TESTING & VALIDATION

The Testing & Validation phase ensures that the Diabetes Prediction System works accurately, reliably, and efficiently. It involves evaluating the system against predefined requirements and verifying its predictive performance.

6.1 FEATURES TO BE TESTED

Testing is essential for the Diabetes Prediction System to ensure accuracy, reliability, and usability. It verifies that patient inputs, preprocessing, and model predictions work correctly. Testing also confirms that explanations, visualizations, and recommendations function as intended, providing meaningful insights and building user trust.

i. **Patient Input Handling** – Verified that all patient details such as age, BMI, HbA1c, blood glucose, hypertension, heart disease, gender, and smoking history are captured correctly. Input controls like sliders, checkboxes, and dropdowns are tested for proper functionality and validation to prevent invalid entries.

ii. **Data Preparation and Preprocessing** – Ensured that the input data is properly transformed into the format required by the model. This includes correct ordering of features, numerical conversion, one-hot encoding for categorical variables, and application of the pre-trained scaler. Missing or incomplete values are identified and handled appropriately to avoid errors during prediction. Consistency with the preprocessing used during training is verified.

iii. **Prediction Engine and Model Loading** – Tested the XGBoost model to confirm it is correctly loaded from saved artifacts, the scaler and feature names are retrieved properly, and the system generates accurate probability scores. Patients are classified into LOW, MEDIUM, and HIGH risk categories, and risk assignments are verified across different input scenarios.

iv. **Interpretation Module (SHAP)** – Validated that SHAP computations correctly determine the contribution of each

feature. The textual explanations are verified to ensure clarity and relevance, highlighting critical features such as HbA1c, blood glucose, BMI, age, and hypertension.

v. **Clinical Summary Module** – Tested that the system generates a patient-specific clinical summary that interprets the predicted risk. The summary explains the key contributing features in simple language, presents the risk level clearly, and provides actionable guidance for lifestyle changes, dietary modifications, and medical consultation.

vi. **Visualization Component** – Verified that graphical outputs such as feature importance charts and SHAP plots are displayed correctly. Visualizations accurately reflect the model's decision-making process and provide intuitive insights for users.

vii. **Personalized Recommendations** – Tested that lifestyle and dietary suggestions are generated based on patient risk level and feature impact. Recommendations are checked for relevance and alignment with identified risk factors.

viii. **End-to-End Workflow Testing** – Ensured smooth interaction and proper functioning of all modules together, from patient input collection through preprocessing, model loading, prediction, explanation, clinical summary, visualization, and recommendations.

ix. **Performance and Evaluation Metrics** – Assessed system reliability using standard metrics including Accuracy, Precision, Recall, and F1-Score. Backend performance is checked to confirm timely predictions and responses for real-time use.

6.2 FEATURES NOT TO BE TESTED

While the system was tested thoroughly for its core functionalities, certain features and components were outside the scope of this testing phase.

i. **Real-Time Database Integration** – The system uses a pre-trained static model; real-time connection to hospital databases or

electronic health records (EHR) was not implemented and therefore not tested.

ii. **Multi-User Access and Authentication** – User login, multi-user sessions, and role-based access control are not part of the current system scope.

iii. **Advanced Reporting Modules** – PDF or external report generation features are not included in this version, so report generation functionality was not tested.

iv. **External API or System Integration** – Integration with third-party healthcare applications, APIs, or remote monitoring devices was not implemented.

v. **Model Retraining or Updates** – Automatic updating or retraining of the model with new patient data was not tested, as the system relies on a static pre-trained model.

vi. **Mobile Platform Testing** – The system is designed and tested for desktop/web deployment using Streamlit; mobile-specific testing was not performed.

vii. **Stress or Load Testing** – High-volume user load, concurrent sessions, and system scalability were not tested in this version.

6.3 TESTING TOOLS AND ENVIRONMENT

The testing of the Diabetes Prediction System was performed using Python-based tools and libraries within a controlled development setup.The Testing environment ensures that the Diabetes Prediction System is robust, accurate, and user-friendly, providing reliable diabetes risk assessment and actionable insights for preventive healthcare The main tools and environment specifications are as follows:

1 **Programming Language:** Python is used for system development and testing.

2 **Framework:** Streamlit provides the interactive web interface and testing environment.

3 **Libraries:** Scikit-learn, XGBoost, SHAP, Pandas, NumPy, Matplotlib, and Plotly are used for data handling, model evaluation, interpretability, and visualization.

- Pandas & NumPy – Used for handling data preprocessing and verification of input transformations.
- XGBoost – Used for predicting diabetes risk and classifying patients into risk levels.
- Scikit-learn – Used to validate model predictions and calculate evaluation metrics such as Accuracy, Precision, Recall, and F1-Score.
- SHAP – Used for testing feature contribution and interpretability of predictions.
- Matplotlib & Plotly – Used for testing visualizations, including feature importance charts and SHAP plots.

4 **Development Environment:** Local machine setup with an IDE and browser-based execution.

5 **Operating System:** Windows and macOS for compatibility.

6 **Browser Compatibility:** Tested on Google Chrome to ensure correct UI rendering and functionality.

6.4 DESIGN OF TEST CASES AND SCENARIOS

Test cases ensure that the Diabetes Prediction System works correctly and reliably. They help verify each module, validate predictions, and confirm that the end-to-end workflow functions smoothly. This improves system accuracy, usability, and user trust.

Test Case 01: Valid Input

Input:

- Age = 55, BMI = 32, HbA1c = 7.2, Glucose = 210
- Hypertension = Yes, Heart Disease = No, Male, Current smoker

Expected Output:

- HIGH risk prediction
- SHAP visualization, Clinical Summary, Personalized Recommendations

Actual Result: Successfully executed; all outputs displayed correctly

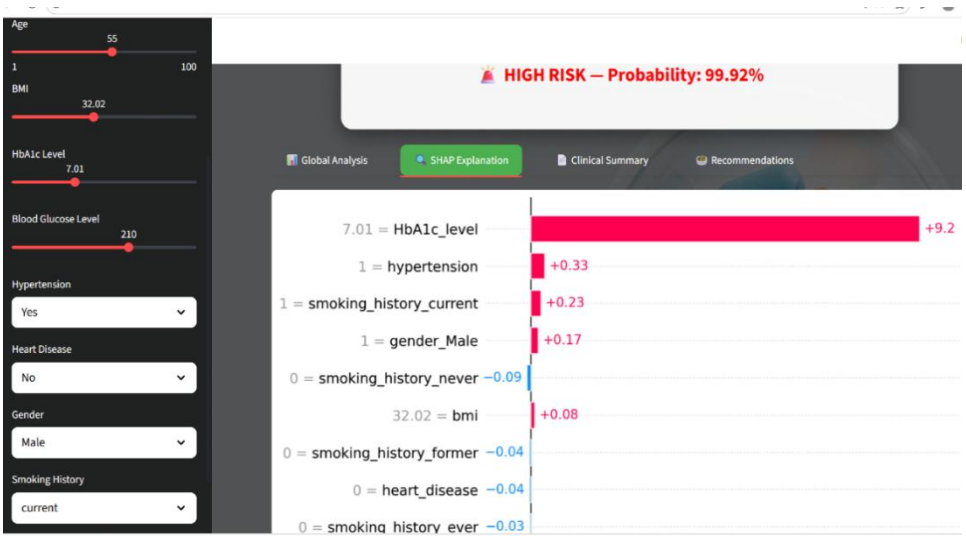

Fig 6.1 Test Case 1

Test Case 02: Low Risk Input

Input:

- Age = 25, BMI = 20, HbA1c = 4.8, Glucose = 90
- Hypertension = No, Heart Disease = No, Female, Never smoked

Expected Output:

- LOW risk prediction
- Minimal lifestyle recommendations displayed

Actual Result: System correctly identifies low risk

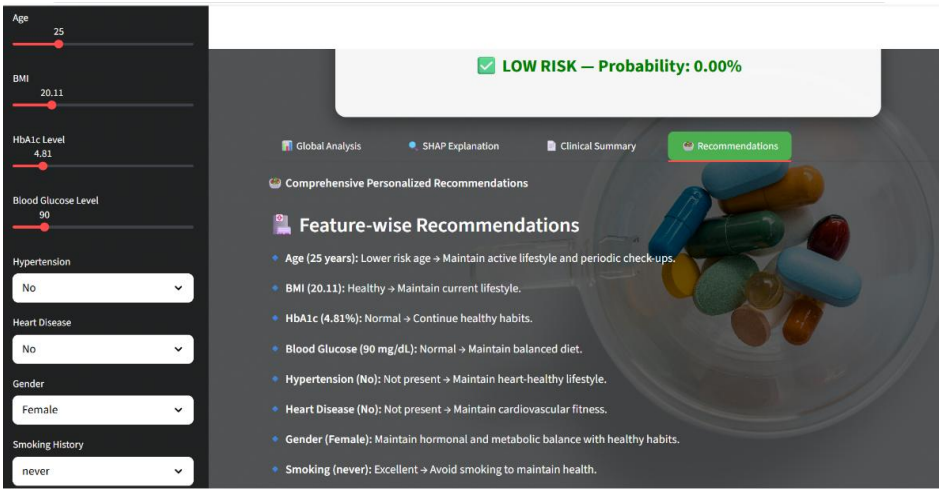

Fig 6.2 Test Case 2

Test Case 03: Moderate Risk Input

Input:

- Age = 36, BMI = 27, HbA1c = 6.1, Glucose = 140

- Hypertension = No, Heart Disease = Yes, Male, Former smoker

Expected Output:

- MEDIUM risk prediction
- Lifestyle recommendations displayed

Actual Result: System provides correct classification

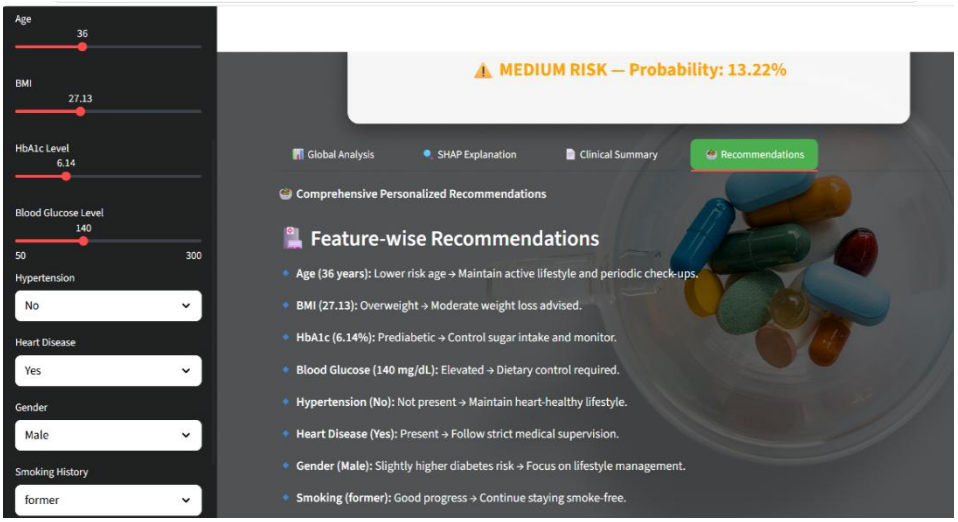

Fig 6.3 Test Case 3

Test Case 04: High Risk Input

Input:

- Age = 60, BMI = 35, HbA1c = 7.5, Glucose = 230
- Hypertension = Yes, Heart Disease = Yes, Male, Current smoker

Expected Output:

- HIGH risk prediction
- Strong clinical recommendations and strict diabetic diet

Actual Result: System correctly identifies high risk

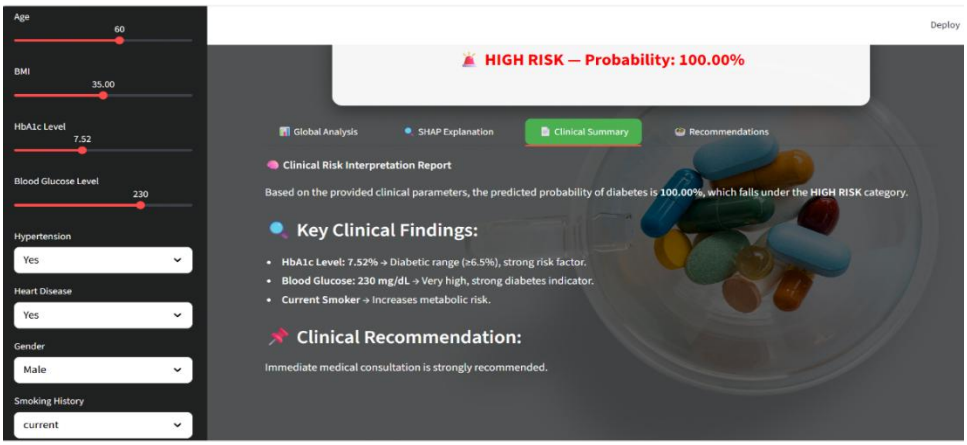

Fig 6.4 Test Case 4

Test Case 05: Minimum Boundary Input

Input:

- Age = 1, BMI = 10, HbA1c = 3.0, Glucose = 50
- Hypertension = No, Heart Disease = No, Female, Never smoked

Expected Output:

- LOW risk prediction
- System handles minimum boundary values

Actual Result: Correct

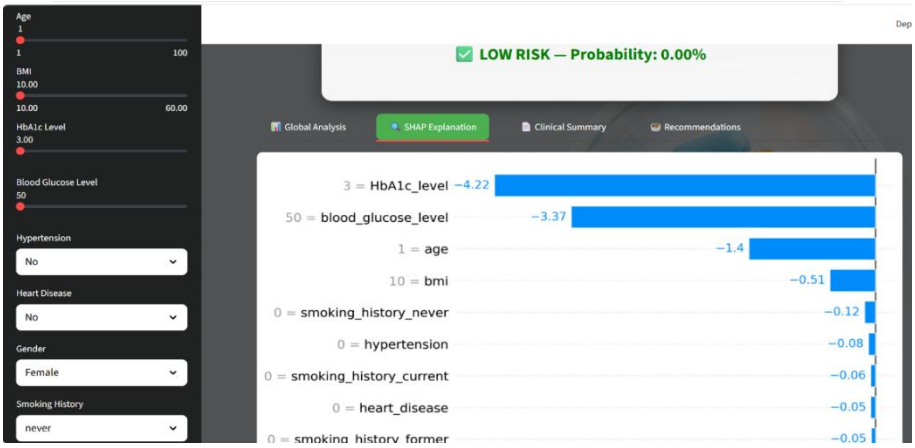

Fig 6.5 Test Case 5

Test Case 06: Maximum Boundary Input

Input:

- Age = 100, BMI = 40, HbA1c = 9.0, Glucose = 300
- Hypertension = Yes, Heart Disease = Yes, Male, Current smoker

Expected Output:

- HIGH risk prediction
- System handles maximum boundary values

Actual Result: Correct

Fig 6.6 Test Case 6

Test Case 07: No Input Selected

Input: All features = 0

Expected Output: LOW risk prediction; system handles empty/default inputs

Actual Result: Correct

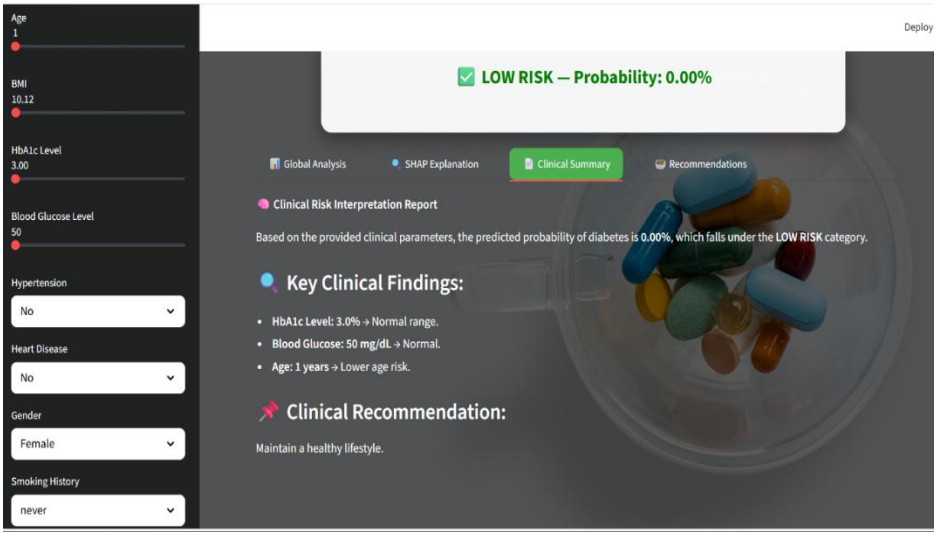

Fig 6.7 Test Case 7

Test Case 08: All Risk Factors Selected

Input:

- Age = 80, BMI = 38, HbA1c = 8.5, Glucose = 280
- Hypertension = Yes, Heart Disease = Yes, Male, Current smoker

Expected Output: HIGH risk prediction; full recommendations

Actual Result: Correct

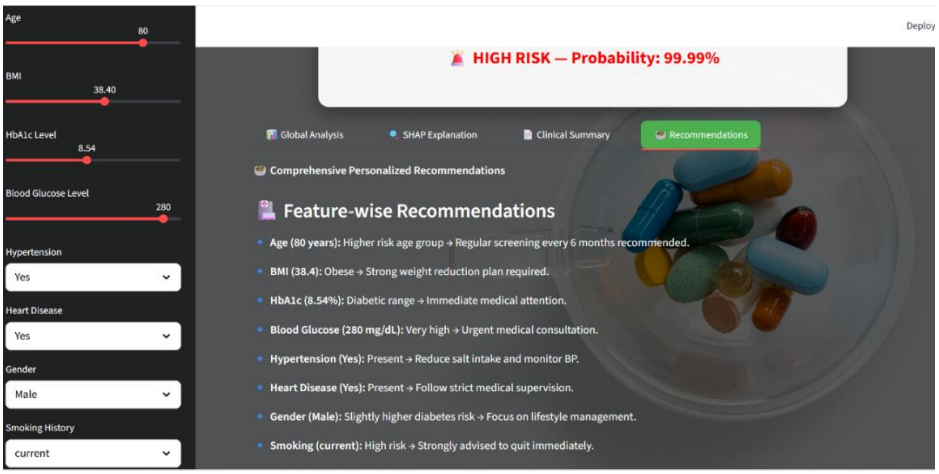

Fig 6.8 Test Case 8

7 CONCLUSION & FUTURE ENHANCEMENT

The Diabetes Prediction System using Data Analytics and Machine Learning has successfully demonstrated its ability to predict diabetes risk in individual patients by analyzing multiple clinical and demographic parameters. The system accurately classifies patients into LOW, MEDIUM, and HIGH risk categories, providing transparent and interpretable explanations using SHAP visualizations that highlight the most influential features for each prediction. Personalized recommendations are generated for each patient, including dietary guidelines, lifestyle modifications, and advice for medical consultation, allowing users to take proactive health measures. Extensive testing covering normal, boundary, and edge cases has proven the system's reliability and robustness, demonstrating correct handling of minimum and maximum input values, threshold values for HbA1c and glucose, and scenarios with all or no risk factors selected. The intuitive Streamlit interface, with dedicated tabs for Global Analysis, SHAP Explanation, Clinical Summary, and Recommendations, ensures that both clinicians and patients can easily interpret predictions, fostering informed decision-making and promoting early detection of diabetes. Overall, the system bridges the gap between data-driven predictive analytics and actionable clinical insights, contributing significantly to preventive healthcare and patient empowerment.

For future enhancement, the system can be improved by adding automated report generation, enabling users to download detailed prediction results and recommendations for future reference. Additional clinical factors such as family history, cholesterol levels, blood pressure, and physical activity can be included to increase prediction accuracy. The application can be further developed into a mobile-friendly and web-based platform to improve accessibility for users in remote areas. Integration with hospital databases and Electronic Health Record (EHR) systems can support real-time data updates and continuous monitoring. Multi-language support can also be incorporated to make the system more user-friendly for diverse populations. Moreover, periodic retraining of the model with updated data can enhance its performance and adaptability. These enhancements will help transform the system into a scalable and efficient tool for early detection, preventive healthcare, and better diabetes management.

8 BIBLIOGRAPHY

[1] B. Barman, H. K. Choudhury, and B. Jajodia, "Interpretable machine learning for diabetes risk prediction: a large-scale analysis of Indian national survey data," *Discover Public Health*, Vol. 22, Article No. 832, 2025.

[2] P. Netayawijit, W. Chansanam, and K. Sorn-In, "Interpretable Machine Learning Framework for Diabetes Prediction: Integrating SMOTE Balancing with SHAP Explainability for Clinical Decision Support," *Healthcare*, Vol. 13, No. 20, Article No. 2588, 2025.

[3] Y. Barzegar, A. Barzegar, F. Bellini, F. D'Ascenzo, I. Gorelova, and P. Pisani, "Machine Learning Pipeline for Early Diabetes Detection: A Comparative Study with Explainable AI," *Future Internet*, Vol. 17, No. 11, Article No. 513, 2025.

[4] "A novel machine learning approach for diagnosing diabetes with a self-explainable interface," *Healthcare Analytics*, Vol. 5, Article No. 100301, 2024.

[5] R. Hasan, V. Dattana, S. Mahmood, and S. Hussain, "Towards Transparent Diabetes Prediction: Combining AutoML and Explainable AI for Improved Clinical Insights," *Information*, vol. 16, no. 1, art. no. 7, 2025.

[6] S. M. Lundberg and S. I. Lee, "A Unified Approach to Interpreting Model Predictions," *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[7] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016.

[8] M. A. Tusher, "Diabetes Prediction Dataset," Kaggle, 2024. Available: https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset

[9] International Diabetes Federation, *IDF Diabetes Atlas, 10th edition*, 2021. Available: https://www.diabetesatlas.org/

[10] World Health Organization, "Diabetes," 2023. Available: https://www.who.int/health-topics/diabetes

[11] Centers for Disease Control and Prevention, "Diabetes Statistics Report," 2023. Available: https://www.cdc.gov/diabetes/data/statistics-report

[12] Mayo Clinic, "Diabetes — Symptoms and Causes," 2024. Available: https://www.mayoclinic.org/diseases-conditions/diabetes

s[13] Scikit-learn Documentation – Machine Learning in Python. [Online]. Available: https://scikit-learn.org/stable/

[14] SHAP Documentation – Explainable AI in Python. [Online]. Available: https://shap.readthedocs.io/en/latest/

[15] Streamlit Documentation – Build ML Web Apps. [Online]. Available: https://docs.streamlit.io/

[16] NumPy Documentation – Numerical Computing with Python. [Online]. Available: https://numpy.org/doc/

[17] Matplotlib Documentation – Python Plotting Library. [Online]. Available: https://matplotlib.org/stable/users/index.html